

Fundamental MATLAB Programming for AI

Today's goals

Module 1

The Fundamentals of MATLAB

Welcome

✓ Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations
Video • 5 min

✓ Specialization and Course Structure
Reading • 5 min

✓ Specialization Sample Certificate
Reading • 1 min

✓ Stay connected with UofG Online
Reading • 5 min

Module 1 Introduction

✓ Introduction to Course 1 and Module 1's Study
Video • 3 min

Module 1-1 MATLAB Online

✓ How to Access MATLAB Online
Reading • 5 min

Module 1-2 MATLAB Basics

✓ MATLAB Basics: The Computing Environment
Video • 11 min

✓ Tutorial: MATLAB Basics
Video • 7 min

✓ Tutorial: MATLAB LiveScript
Video • 7 min

✓ MATLAB Basics: Using Built-in Functions
Video • 3 min

✓ Materials about MATLAB Basics
Reading • 10 min

Module 1-3 MATLAB Matrices

✓ MATLAB Matrices: Enter a Matrix
Video • 8 min

✓ MATLAB Matrices: Access Elements in a Matrix
Video • 5 min

✓ MATLAB Matrices: Manipulate Matrices
Video • 7 min

✓ Materials about MATLAB Matrices
Reading • 10 min

Module 1-4 MATLAB Built-in Functions

✓ MATLAB Built-in Functions: Data Types, Matrix Operations, and Statistics
Video • 9 min

✓ MATLAB Built-in Functions: Matrix Tiling, Sorting, and Random Number Generation
Video • 6 min

✓ Materials about MATLAB Built-in Functions
Reading • 10 min

Module 1-5 MATLAB Scripts

✓ MATLAB Scripts: Introduction to MATLAB Scripts
Video • 5 min

✓ Tutorial: MATLAB Debug Example
Video • 9 min

✓ Materials about MATLAB Scripts
Reading • 10 min

Module 1-6 MATLAB Graphics

✓ MATLAB Graphics: Plotting Basics and Customization
Video • 5 min

 MATLAB Graphics: Improving the Plots

Module 1 Recap

 0:07 / 1:22 

2x





You have completed the first module of this course. At this point, you should be able to:

- Work confidently with the MATLAB environment and workspace
- Create and manipulate matrices and vectors
- Use built-in functions for computation and data processing
- Write and debug simple scripts
- Visualize data using plots and customized figures

These are not just programming basics. They are the computational foundations for everything that follows in this specialization. Most AI algorithms operate on matrices, rely heavily on vectorized operations, and require careful implementation and validation. If you are comfortable working with data, functions, and visualizations in MATLAB, you are already building the core skills needed for applied AI.

Before moving on, make sure you can:

- Predict the size and structure of your outputs before running code
- Use element-wise operations correctly ($\ast$, $\./$, $\.^{\wedge}$)
- Debug unexpected results by checking intermediate variables and dimensions

In the next module, you will move beyond basic scripts and start writing **structured programs**. You will learn how to control program flow using **conditionals and loops**, and how to create **custom functions**. These skills will allow you to build reusable, scalable solutions for real engineering and scientific problems.

You are now moving from using MATLAB as a calculator to using it as a programming tool.

Go to next item

✓ Completed

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